4. Quality of water

The cost of the treatment depends on the quality of water. Bad quality water requires excessive treatment and increases the cost. But the good quality requires less treatment and decreases the cost of the scheme.

The sources from which water is obtained for water supply can be classified into two types

1. Surface sources
2. Sub surface sources

4.2.1 Surface sources

The primary source of water is rain. When rain falls on the ground, a certain portion percolates into the ground and the balance portion remains on the ground as surface water. The usual forms of surface sources are as follows:

i. Ponds
ii. Lakes and streams
iii. Storage reservoirs
iv. Rivers
v. Sea

4.2.2 Sub surface sources

These sources obtain their supply from percolation of precipitation. It is comparatively safer, free from suspended impurities and are reliable. It is due to the fact that water get strained during their passage through the porous underground strata. This water contains more dissolved mineral and gases. The bacterial content is usually low. In General, sub surface source water is good in quality. But they may require some treatment to improve their chemical characteristics.

The usual forms of subsurface sources are as follows:

i. Springs
ii. Wells
iii. Infiltration galleries
iv. Infiltration wells



Well

Activity 1

Identify the sources of water in your school and home. Brief two points of each.

4.3 Quality of Water

The water required for water supplies should be pure. It should be free from pathogens, toxic chemicals, etc., in order to provide good quality. The water should be pre-treated before distribution to the consumers.

Wholesome water

Water which does not contain harmful impurities and which are good or harmless to health is termed as wholesome water. It is bacteriologically pure but chemically not pure.

Distilled water

Distilled water is safe water as it is bacteriologically and chemically pure. But it is unpleasant to drink.

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